

Project 17: JaniKing: Online Commercial Cleaning Franchise Portal

About the Client

JaniKing have been backing successful commercial cleaning franchises for over 30 years and one of the keys to our success is the idea that owner-operators deliver the very best service because they have a personal stake in their business.

A Jani-King commercial cleaning franchise will introduce you to a world of business excellence and significant opportunities, with one of the best perks being that we welcome individuals with no prior business or commercial cleaning experience to come on board.

Jani-King provides cleaning solutions to a range of Industries, including offices, education, retail, healthcare, industrial and hospitality. Because no two businesses are the same, we provide a comprehensive and fully tailored cleaning solution for every client we serve, with thousands of businesses across Australia trusting Jani-king as their cleaning provider.

We are committed to offering our clients value for money and most importantly, a comprehensive and thoroughly checked cleaning service. With 24/7 support for clients and cleaning franchises, you can rest assured you will be looked after from the moment you choose Jani-King.

Our commercial cleaners use the most technologically advanced products, equipment, techniques and systems and we are committed to providing all clients with safe, sustainable and responsible cleaning services.

Project Brief & Business Problem Specifications:

Manual Franchisee Onboarding

- Complex, paper-based application and approval process.
- Delays in verifying documents and tracking statuses.

Decentralised Data Management

- No central system for franchisee records.
- Difficulty accessing up-to-date contact and store information.

Inconsistent Product and Service Catalogues

- Franchisees using outdated or incorrect product details.
- Lack of standard pricing and service offerings.

Poor Order and Inventory Visibility

- HQ can't track stock levels across locations.
- Risk of stockouts or overstocking without alerts.

Payment and Billing Challenges

- Manual invoicing processes lead to errors.
- Hard to track outstanding franchise payments.

Limited Communication Channels

- Reliance on email chains and phone calls.
- No structured messaging or announcement system.

Weak Sales and Performance Tracking

- No centralised view of sales and revenue data.
- Hard to compare franchisee performance.

Uncoordinated Marketing Efforts

- Inconsistent local promotions.
- Difficult to plan, launch, and monitor campaigns.

Inefficient Support Processes

- Franchisees struggle to get timely help.
- No standard support ticketing system.

System Modules Requirements

- User and Access Management Module
- Franchise Management Module
- Client and Location Management Module
- Cleaning Service Management Module
- Service Request and Quotation Module
- Cleaning Contract Management Module

- Booking and Scheduling Module
- Staff and Timesheet Management Module
- Cleaning Checklist and Inspection Module
- Equipment and Inventory Management Module
- Invoice and Payment Management Module
- Franchise Fee and Royalty Management Module
- Complaint, Incident and Notification Module
- Dashboard, Reporting and Administration Module

User Modules (User Frontend):

Developers need to research and discuss with the client to finalise the modules and requirements.

UI Design

User Interface Design is concerned with the dialogue between a user and the computer. It is concerned with everything from starting the system or logging into the system to the eventual presentation of desired inputs and outputs. The overall flow of screens and messages is called a dialogue.

UI Design Requirements

1. The system user should always be aware of what to do next.
2. The screen should be formatted so that various types of information, instructions and messages always appear in the same general display area.
3. Message, instructions or information should be displayed long enough

to allow the system user to read them.

4. Use display attributes sparingly.
5. Default values for fields and answers to be entered by the user should be specified.
6. A user should not be allowed to proceed without correcting an error.
7. The system user should never get an operating system message or fatal error.

The aim of proposed system is to develop a system of improved facilities. The proposed system can overcome all the limitations of the existing system. The system provides proper security and reduces the manual work.

- Security of data.
- Ensure data accuracy's
- Proper control of the higher officials.
- Minimize manual data entry.
- Minimum time needed for the various processing.
- Greater efficiency.
- Better service.
- User friendliness and interactive.
- Minimum time required.

Functional Requirements

Functional requirements are product features or functions that developers must implement to enable users to accomplish their tasks. So, it's important to make them clear for the stakeholders. Generally, functional requirements describe system behavior under specific conditions. The developers of this system must enhance the performance and efficiency of the system by adding all the functional requirements. Students need to do their own research to find how they can improve the system and which FRs need to added. The group must need a prior approval from the stakeholders/project supervisor before finalizing these Functional Requirements. These enhanced FRs must be reflected separately in Final SRS Report after the approval.

Module 1: User and Access Management

- FR01.** The system should allow authorised administrators to create user accounts.
- FR02.** The system should allow users to log in using a registered email address and password.
- FR03.** The system should allow users to securely log out of the portal.
- FR04.** The system should assign access permissions according to user roles, including administrator, franchisor, franchisee, supervisor, cleaning staff, finance officer and client.
- FR05.** The system should allow users to reset forgotten passwords through their registered email addresses.

Module 2: Franchise Management

- FR06.** The system should allow the franchisor to register a new cleaning franchise.
- FR07.** The system should store the franchise name, franchise owner, contact details, business registration details and commencement date.
- FR08.** The system should allow the franchisor to approve, reject, suspend or terminate a franchise account.
- FR09.** The system should allow service territories and postcode areas to be assigned to each franchise.
- FR10.** The system should maintain franchise agreement commencement, expiry and renewal details.

Module 3: Client and Location Management

- FR11.** The system should allow franchisees to register commercial clients.
- FR12.** The system should store client business details, contact person, email address, telephone number and billing address.
- FR13.** The system should allow multiple service locations to be registered under one client account.
- FR14.** The system should allow users to search for clients by business name, contact person, account number or location.
- FR15.** The system should display the service, quotation, booking, invoice and complaint history of each client.

Module 4: Cleaning Service Management

- FR16.** The system should allow administrators to create cleaning service categories.
- FR17.** The system should store the service name, description, standard price, estimated duration and required number of staff.
- FR18.** The system should support services such as office cleaning, carpet cleaning, window cleaning, floor maintenance and sanitisation.
- FR19.** The system should allow franchisees to adjust service prices within limits

established by the franchisor.

FR20. The system should allow cleaning services to be activated, updated or deactivated.

Module 5: Service Request and Quotation

FR21. The system should allow clients to submit commercial cleaning service requests online.

FR22. The system should collect the requested service, location, preferred date, property size, cleaning frequency and special instructions.

FR23. The system should automatically assign a service request to the franchise responsible for the client's area.

FR24. The system should allow franchisees to prepare quotations containing labour costs, material costs, discounts, taxes and additional charges.

FR25. The system should allow clients to accept or reject quotations through the portal.

Module 6: Cleaning Contract Management

FR26. The system should allow accepted quotations to be converted into cleaning contracts.

FR27. The system should generate a unique contract reference number.

FR28. The system should store the contract start date, end date, service frequency, service price, scope of work and payment terms.

FR29. The system should allow authorised users to upload signed contract documents.

FR30. The system should allow contracts to be renewed, suspended, amended or terminated.

Module 7: Booking and Scheduling

FR31. The system should allow franchisees to create cleaning service bookings.

FR32. The system should support one-time, daily, weekly, fortnightly, monthly and custom recurring bookings.

FR33. The system should display cleaning bookings in daily, weekly and monthly calendar views.

FR34. The system should prevent the same staff member from being assigned to overlapping bookings.

FR35. The system should allow authorised users to reschedule or cancel bookings and record the reason for the change.

Module 8: Staff and Timesheet Management

FR36. The system should allow franchisees to register cleaning staff and supervisors.

FR37. The system should store staff contact details, employment status, qualifications, training records and availability.

FR38. The system should allow supervisors to assign one or more staff members to a cleaning booking.

FR39. The system should allow staff members to clock in and clock out for assigned cleaning jobs.

FR40. The system should allow supervisors to review, approve, reject or correct staff timesheets.

Module 9: Cleaning Checklist and Inspection

FR41. The system should allow supervisors to create cleaning checklist templates.

FR42. The system should allow different cleaning checklists to be assigned to different services and client locations.

FR43. The system should allow cleaning staff to mark checklist tasks as completed during a cleaning job.

FR44. The system should allow supervisors to perform quality inspections and record ratings, comments and photographs.

FR45. The system should generate corrective action tasks when a cleaning inspection receives an unsatisfactory result.

Module 10: Equipment and Inventory Management

FR46. The system should allow franchisees to register cleaning equipment.

FR47. The system should store equipment identification numbers, purchase dates, conditions, assigned locations and maintenance dates.

FR48. The system should allow users to record equipment faults, repairs and maintenance activities.

FR49. The system should allow franchisees to record cleaning supply stock received, used, damaged or disposed of.

FR50. The system should generate low-stock and equipment maintenance alerts.

Module 11: Invoice and Payment Management

FR51. The system should allow franchisees to generate invoices for completed cleaning services.

FR52. The system should calculate service charges, discounts, additional charges, taxes and the total amount payable.

FR53. The system should generate a unique invoice number for every invoice.

FR54. The system should record invoice statuses, including draft, issued, partially paid, paid, overdue and cancelled.

FR55. The system should allow authorised users to record payments and generate payment receipts.

Module 12: Franchise Fee and Royalty Management

FR56. The system should allow the franchisor to define franchise fees and

royalty rates.

FR57. The system should calculate royalty amounts based on franchise revenue or a predefined fee structure.

FR58. The system should generate franchise fee and royalty statements.

FR59. The system should allow franchisees to view their royalty payment history and outstanding balances.

FR60. The system should allow finance officers to record and update franchise fee and royalty payments.

Module 13: Complaint, Incident and Notification

FR61. The system should allow clients to submit complaints through the portal.

FR62. The system should allow staff members to report workplace injuries, property damage, safety hazards and security incidents.

FR63. The system should assign complaints and incidents to an authorised franchisee, supervisor or administrator.

FR64. The system should record actions taken, responsible users, progress notes and resolution details for each complaint or incident.

FR65. The system should send notifications for bookings, schedule changes, invoices, overdue payments, complaints, inspections and expiring documents.

Module 14: Dashboard, Reporting and Administration

FR66. The system should provide a personalised dashboard based on the logged-in user's role.

FR67. The system should generate reports for bookings, revenue, payments, staff performance, inspections, complaints and franchise performance.

FR68. The system should allow reports to be filtered by date, franchise, client, service, location and status.

FR69. The system should allow reports to be exported in PDF or CSV format.

FR70. The system should maintain an audit log containing the user, activity, affected record, date and time of important system actions.

Non-Functional Requirements

NFR1. Performance

- Ensure average page load times under 2 seconds.
- Optimise PHP queries and MySQL indexing for fast response.

NFR2. Scalability

- Design database schemas for growing numbers of franchises, bookings, and staff.

- Support horizontal scaling of PHP servers if needed.

NFR3. Availability

- Maintain at least 99.5% uptime excluding planned maintenance.
- Implement redundant server setups where required.

NFR4. Reliability

- Guarantee reliable booking, payment, and notification processes without data loss.
- Use ACID-compliant MySQL transactions for critical operations.

NFR5. Maintainability

- Follow PSR-12 PHP coding standards for clear, consistent code.
- Use modular PHP architecture to allow easy updates and feature additions.

NFR6. Security – Data Storage

- Hash all user passwords securely (e.g., bcrypt).
- Encrypt sensitive data in the MySQL database.

NFR7. Security – Data Transmission

- Enforce HTTPS for all client-server communication.
- Protect customer and franchisee personal data in transit.

NFR8. SQL Injection Protection

- Use prepared statements or ORM to prevent SQL injection attacks.
- Validate all user inputs on both client and server sides.

NFR9. Session Management

- Implement secure PHP session handling with timeouts.
- Regenerate sessions on login to prevent hijacking.

NFR10. Access Control

- Enforce role-based permissions (Admin, Franchisee, Staff) throughout the portal.
- Restrict access to sensitive features and data based on roles.

NFR11. Audit Logging

- Log critical user actions such as logins, bookings, payments, approvals.
- Store logs in MySQL with timestamps for auditing and compliance.

NFR12. Backup and Recovery

- Perform daily automated MySQL database backups.
- Enable full restoration within 4 hours of a critical failure.

NFR13. Usability

- Provide an intuitive, user-friendly interface for all roles.
- Include clear navigation, labels, and help prompts.

NFR14. Responsiveness

- Design pages to work smoothly on desktop, tablet, and mobile devices.
- Use responsive HTML/CSS frameworks for consistent layout.

NFR15. Compatibility

- Support modern browsers (Chrome, Firefox, Edge, Safari).
- Ensure compatibility with PHP 7.4+ and MySQL 5.7+ or higher.

NFR16. Accessibility

- Follow WCAG 2.1 Level AA guidelines where feasible.
- Include keyboard navigation and screen-reader-friendly layouts.

NFR17. Notification Reliability

- Ensure email/SMS notifications are reliably sent using PHP integrations.
- Log all notifications and failures in MySQL for review.

NFR18. Reporting Accuracy

- Generate accurate PDF and CSV reports on bookings, payments, and franchise activity.
- Use consistent data queries with validation for accuracy.

NFR19. Payment Security

- Integrate with PCI-compliant payment gateways.
- Avoid storing raw credit card data in MySQL.

Hardware Requirement: Should be recommended by the developers.

Software Requirement: Should be recommended by the developers.